

Alanine Aminotransferase (ALT/SGPT)

Test Details

METHODOLOGY

Kinetic

RESULT TURNAROUND TIME

Within 1 day

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Use

A liver function test, ALT is more sensitive for the detection of hepatocyte injury than for biliary obstruction. ALT is more specific for liver injury than AST (SGOT). Useful for hepatic cirrhosis, other liver disease. Increased in Reye syndrome, with AST.¹ Test for hepatitis. Acute hepatitis A or B can be confirmed serologically, as can hepatitis C. Negative serological findings in the presence of hepatitis-like chemistry abnormalities may also suggest acute drug-induced hepatitis, an impression supported by resolution after removal of the offending agent.² The combination of increased AST and ALT with negative hepatitis markers occurs in a number of other entities including infectious mononucleosis. Sensitive to heart failure.

Limitations

Grossly hemolyzed samples can generate somewhat spurious results. The activity in red cells is six times that of serum. Elevations are reported in trauma to striated muscle, rhabdomyolysis, polymyositis and dermatomyositis, but the CK (CK-MM fraction) is increased in such patients and it is preferable to consider diseases of skeletal muscle. ALT is less sensitive than is AST to alcoholic liver disease. Increased ALT is found with obesity.

Custom Additional Information

Among entities in which AST and ALT increases occur, are therapeutic applications of bovine or porcine heparin. LD (LDH) abnormality with elevation of hepatic fractions was also reported.³ In children with acute lymphoblastic leukemia, high ALT activity at diagnosis is associated with rapidly progressive ALL.⁴ A number of drugs, including diphenylhydantoin, heparin therapy and many others, cause ALT increases. Acetaminophen hepatotoxicity may be potentiated in alcoholics, in whom coagulopathy and extremely abnormal aminotransferase levels are described, ALT less than AST.⁵ The hepatitis C virion has been detected by polymerase chain reaction and reverse transcriptase of HCV-RNA sequences in patients with elevated ALT and positive anti-HCV.⁶

Specimen Requirements

Specimen

Serum (preferred) **or** plasma

Volume

1 mL

Minimum Volume

0.7 mL (**Note:** This volume does **not** allow for repeat testing.)

Container

Red-top tube, gel-barrier tube, **or** green-top (lithium heparin) tube. Do **not** use oxalate, EDTA, or citrate plasma.

Collection Instructions

Separate serum or plasma from cells within 45 minutes of collection.

Stability Requirements

Temperature	Period
Room temperature	7 days
Refrigerated	14 days

Storage Instructions

Maintain specimen at room temperature.

Causes for Rejection

Gross hemolysis; excessive lipemia; improper labeling

References

Tonks DB. A study of the accuracy and precision of clinical chemistry determinations in 170 Canadian laboratories. *Clin Chem*. 1963 Apr; 9:217-233. PubMed 13985504 (<http://www.ncbi.nlm.nih.gov/pubmed/13985504>)

Footnotes

1. Diagnosis and treatment of Reye's syndrome. *JAMA*. 1981 Nov 27; 246(21):2441-2444. PubMed 7299966 (<http://www.ncbi.nlm.nih.gov/pubmed/7299966>)

2. Frank BB. Clinical evaluation of jaundice. A guideline of the Patient Care Committee of the American Gastroenterological Association. *JAMA*. 1989 Dec 1; 262(21):3031-3034. PubMed 2681857

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3. Dukes GE Jr, Sanders SW, Russo J Jr, et al. Transaminase elevations in patients receiving bovine or porcine heparin. *Ann Intern Med*. 1984 May; 100(5):646-650. PubMed 6712030 (<http://www.ncbi.nlm.nih.gov/pubmed/6712030>)

4. Rautonen J, Siimes MA. Elevated serum transaminase activity at diagnosis is associated with rapidly progressing disease in children with acute lymphoblastic leukemia. *Cancer*. 1988 Feb 15; 61(4):754-757.

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5. Seeff LB, Cuccherini BA, Zimmerman HJ, Adler E, Benjamin SB. Acetaminophen hepatotoxicity in alcoholics. A therapeutic misadventure. *Ann Intern Med*. 1986 Mar; 104(3):399-404. PubMed 3511825

(<http://www.ncbi.nlm.nih.gov/pubmed/3511825>)

6. Ulrich PP, Romeo JM, Lane PK, Kelly I, Daniel LJ, Vyas GN. Detection, semiquantitation, and genetic variation in hepatitis C virus sequences amplified from the plasma of blood donors with elevated alanine aminotransferase. *J Clin Invest*. 1990 Nov; 86(5):1609-6914. PubMed

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